Kernighan–Lin bipartition

Kernighan–Lin improves an existing bipartition by swapping pairs across the cut

The idea

- Score each unlocked pair by swap gain
- Here the winning prefix is one swap: A with D, gain nine
- Pass one then finds nothing and KL stops at a local optimum for swap moves
- <!-- algorithm-walkthrough -->

Bad seed: cutsize 12

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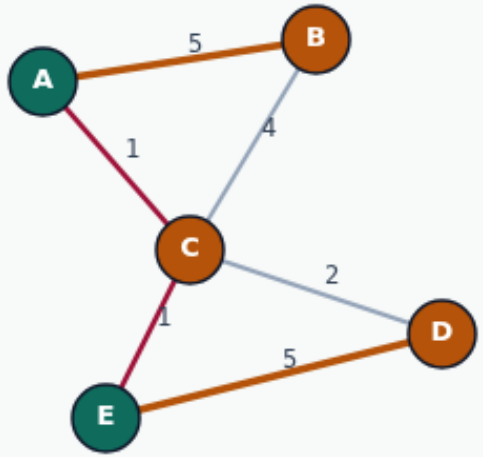
Bad seed: cutsize 12

Step 1 / 5 · bad-seed · module02-01-kl-partition



Explanation

Start from a terrible bipartition AE|BCD. Both heavy edges A–B and D–E are cut, so cutsize is 12. KL will search improving swaps.

- Seed parts: A,E vs B,C,D
- Cut includes A–B(5) and D–E(5)
- Goal: reduce cut without enumerating all partitions

```
seed:
{"A":"0","E":"0","B":"1","C":"1","D":"1"}
cutsize: 12
```

Bad seed: cutsize 12

Score pairwise swaps by gain

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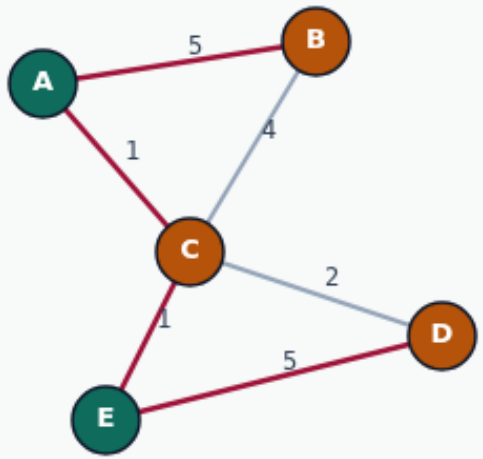
Score pairwise swaps by gain

Step 2 / 5 · gain-idea · module02-01-kl-partition



Explanation

KL considers swapping one vertex from each side. Gain estimates how much the cut shrinks. The best unlocked pair here is A ↔ D with gain 9.

- D(v)=external−internal for each vertex
- Swap gain uses D values and the edge between the pair
- Lock pairs after considering them in a pass

```
Best candidate swap: A ↔ D
gain g = 9
```

Score pairwise swaps by gain

Accept prefix: only A ↔ D

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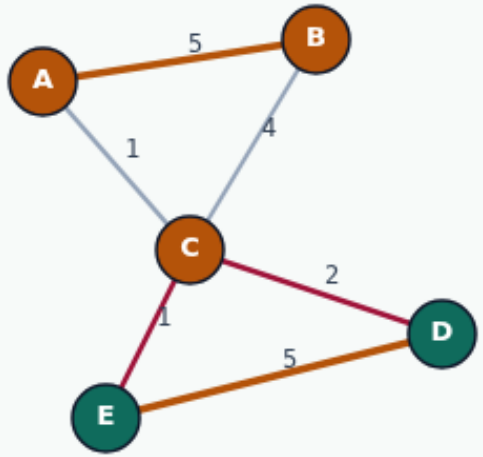
Accept prefix: only A ↔ D

Step 3 / 5 · accept-swap · module02-01-kl-partition



Explanation

Pass 0 builds a sequence of candidate swaps, then keeps the prefix with best cumulative gain. Here best_k=1: perform A ↔ D once.

- best_k = 1
- bestCum = 9
- cut 12 → 3 in one swap

```
pass 0: A/D(9)
cutBefore=12 cutAfter=3
improved=true
```

Accept prefix: only A ↔ D

Refined partition ABC|DE

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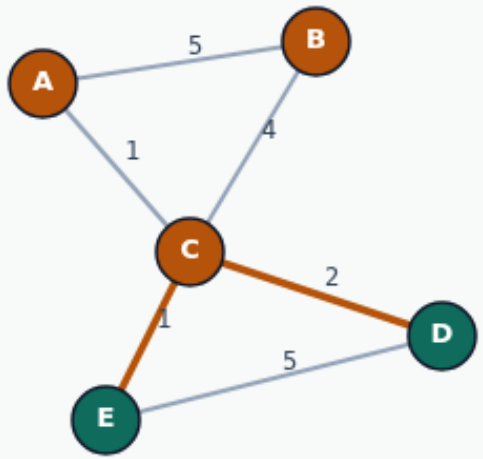
Refined partition ABC|DE

Step 4 / 5 · final · module02-01-kl-partition



Explanation

After the swap, A joins B,C and D joins E. Heavy edges are internal; only the weak C–D/C–E bridge remains cut.

- Final parts: ABC|DE
- Matches the greedy/LP communities
- Cutsize golden: 3

```
cutsize: 3
parts: ABC|DE
```

Refined partition ABC|DE

Next pass finds nothing

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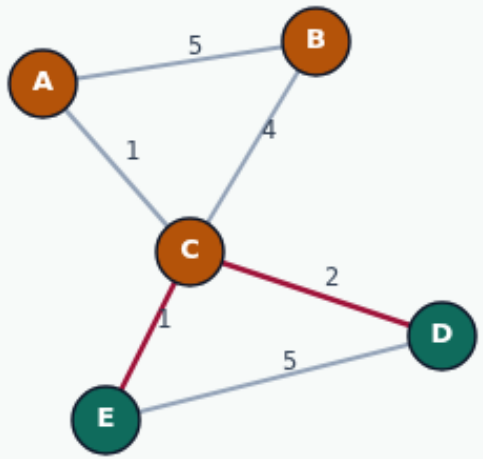
Next pass finds nothing

Step 5 / 5 · pass1-stop · module02-01-kl-partition



Explanation

Pass 1 reports improved=false. KL stops when a full pass cannot improve — local optimum for swap moves from this seed.

- Local, not global, optimum
- Quality depends on the seed
- Still the classic bipartition refiner

```
pass 1: best_k=0 improved=false
stop
```

Next pass finds nothing

Browser lab track

- In the browser lab track, open the **kl-partition** lab from the tools shelf
- Load the starter graph, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module’s examples and the course `common/` solvers
- Parse the tiny graph, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps
- For multilevel flows, verify coarsening before you blame the refiner

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you’re ready, take the short quiz, then continue to the next module